

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	5 September 2026
Team ID	SWUID2026190411
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Station Selection	Select Starting From station Select Going To station
FR-2	Journey Selection	Select Single Journey or Return Journey
FR-3	Passenger Selection	Select number of passengers for the ticket
FR-4	Fare Calculation	Automatically calculate fare based on journey and passenger details
FR-5	Payment Selection	Select payment mode: UPI, Card, or Others
FR-6	Ticket Booking	Submit metro ticket booking request through the ServiceNow Catalog Item
FR-7	QR Ticket Generation	Generate and display a QR code for the metro ticket
FR-8	Station Details Management	Store and maintain metro station details in the Station Details table
FR-9	Ticket Record Management	Create and maintain ticket/request information in ServiceNow
FR-10	Access Control	Allow authorized users to perform permitted operations on project records

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The ticket booking form should be simple and easy to use, allowing passengers to select stations, journey type, passengers, payment mode, and submit the request with minimal steps.
NFR-2	Security	ServiceNow access controls should protect station and ticket-related records and restrict operations to authorized users.
NFR-3	Reliability	The system should consistently calculate fares, process ticket requests, and display the generated QR ticket without unexpected failures.
NFR-4	Performance	The booking form, fare calculation, and QR display should respond quickly to provide a smooth ticket-booking experience.
NFR-5	Availability	The solution should be available through the ServiceNow platform whenever the ServiceNow instance and required services are operational.
NFR-6	Scalability	The solution should support increasing numbers of stations, passengers, ticket requests, and transaction records without requiring major changes to the basic design.